Regulation Landscape

CEC 300: Computing in Aviation and Aerospace

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FAA Certification and Standards

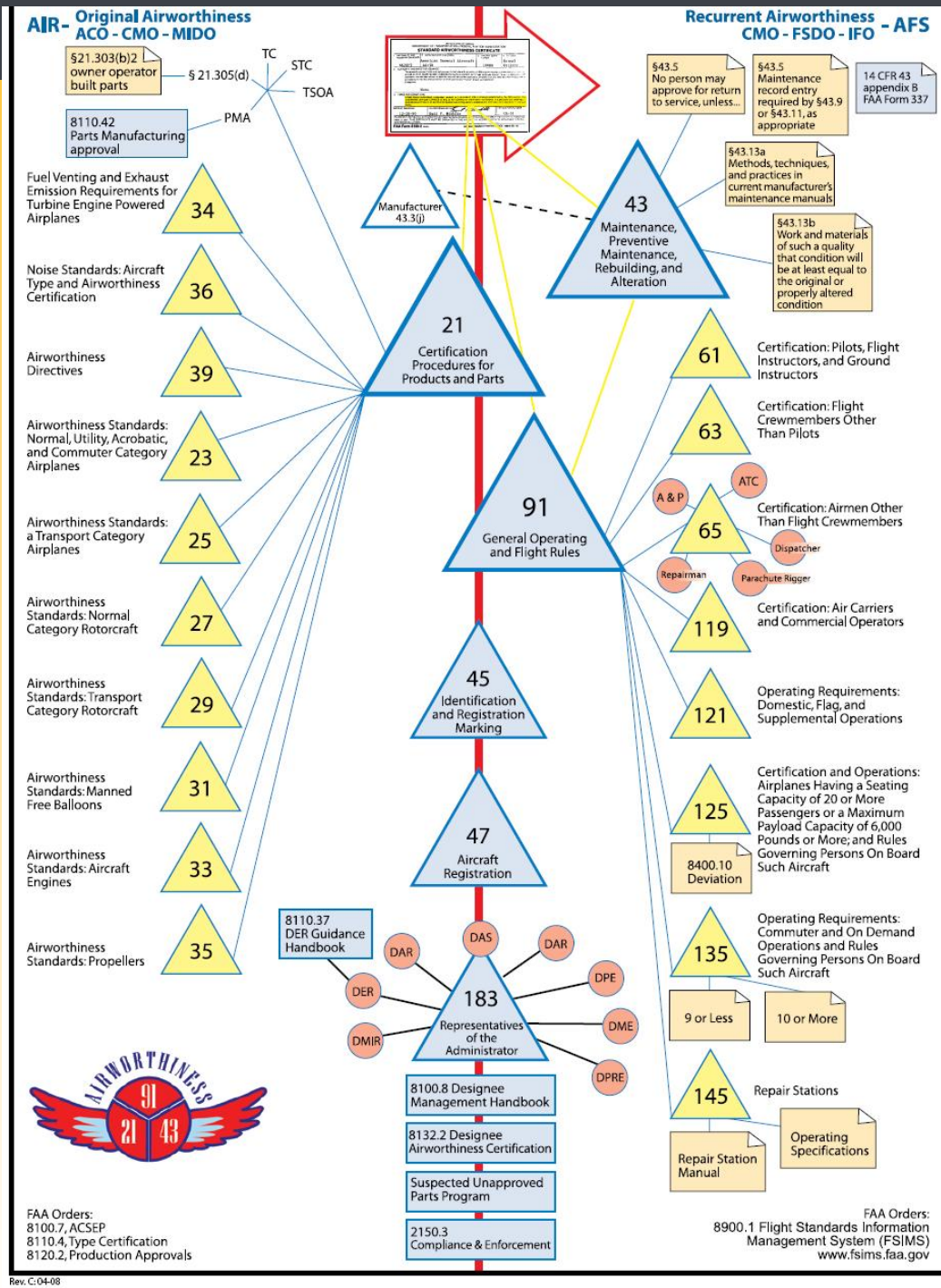

Figure 13-5. Graphic chart of FAA regulations

Terminology

- **Airworthiness Certificate:** a certificate stating an aircraft has been deemed airworthy, which includes conformance to its type certificate and continued airworthiness (maintenance, repair, etc.)
    - Standard
    - Special: primary, restricted, limited, light-sport, and experimental (i.e., R&D, crew training, exhibition, marketing)

- **Type Certificate (TC):** a certificate acknowledging airworthiness compliance for an aircraft type.

- **Production Certificate:** a certificate authorizing production of aircraft based upon a type certificate.

- **Special Conditions:** can be designated by the FAA for an aircraft type that does not conform to existing airworthiness standards of 14 CFR because of novel / unique design features.

Terminology

- **Airworthiness Directive (AD):**
  - "Airworthiness Directives (ADs) are legally enforceable rules issued by the FAA in accordance with 14 CFR part 39 to correct an unsafe condition in a product. 14 CFR part 39 defines a product as an aircraft, aircraft engine, propeller, or appliance." (FAA)
  - Responsibility of aircraft owner / operator to take action

- **Advisory Circular (AC):**
  - FAA published documents that provide guidance regarding acceptable means of compliance for FAA standards such as airworthiness
  - Not binding and not the sole means of compliance
  - Sometimes reference industry-base standards (RTCA, ASTM, etc.)

- **Technical Standard Order (TSO):**
  - "A TSO is a minimum performance standard for specified materials, parts, and appliances used on civil aircraft. When authorized to manufacture a material, part, or appliances to a TSO standard, this is referred to as TSO authorization. Receiving a TSO authorization is both design and production approval."
  - A manufacturer can obtain a TSO authorization for a product indicating that their product meets the standards and is authorized to produce the product

- **Orders:**
  - Documents defining FAA policy and procedures for FAA personnel and representatives
  - Sometimes the orders can provide guidance to developers, operators, etc. on how regulation is interpreted and compliance is assessed.

Aviation Regulations

- FAA is part of the executive branch
  - Given delegated authority from Congress, the FAA establishes regulatory law.
  - Regulatory laws in the U.S. Government are available to the public through the **Codes of Federal Regulations (CFR)**

- **Title 14** contains regulations related to Aeronautics and Space

- Aviation regulations established by the FAA are found in Chapter 1.
  - Chapters are organized in subchapters

- Chapters have subchapters.

- Subchapters have Parts.

- Parts include rules that can be further divided into subparts and paragraphs.

ECFR CONTENT

		Part / Section
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▶ **Chapter V**	National Aeronautics and Space Administration	1200 – 1299
▶ **Chapter VI**	Air Transportation System Stabilization	1300 – 1399

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▶ **Part 26**	Continued Airworthiness and Safety Improvements for Transport Category Airplanes	26.1 – 26.49
▶ **Part 27**	Airworthiness Standards: Normal Category Rotorcraft	27.1 – 27.1589
▶ **Part 29**	Airworthiness Standards: Transport Category Rotorcraft	29.1 – 29.1589
▶ **Part 31**	Airworthiness Standards: Manned Free Balloons	31.1 – 31.85
▶ **Part 33**	Airworthiness Standards: Aircraft Engines	33.1 – 33.201
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▶ **Part 36**	Noise Standards: Aircraft Type and Airworthiness Certification	36.0 – 36.1583
▶ **Part 38**	Airplane Fuel Efficiency Certification	38.1 – 38.23
▶ **Part 39**	Airworthiness Directives	39.1 – 39.27
▶ **Part 43**	Maintenance, Preventive Maintenance, Rebuilding, and Alteration	43.1 – 43.17
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Subpart A	General	48.1 – 48.30
Subpart B	Certificates of Aircraft Registration for Small Unmanned Aircraft	48.100 – 48.125
Subpart C	Aircraft Marking	48.200 – 48.205
▶ **Part 49**	Recording of Aircraft Titles and Security Documents	49.1 – 49.63

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Aviation Regulations

- Generating new laws can be cumbersome.

- General Process:
  - Aviation rulemaking committees (ARCs) can assist in developing new rules
  - A draft rule must be vetted by FAA and DOT
  - A notice of public rulemaking is issued
  - A period of public comment occurs
  - FAA must address all comments
  - Revision is made toward final regulation
  - Approval through FAA and DOT before becoming a new regulatory law.

- Example: Current Beyond Visual Line of Sight UAS Operations just finished public comment period. If approved, it becomes Part 108.

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14 CFR Part 21

▼ **Part 21**	Certification Procedures for Products and Articles	21.1 – 21.700
	Special Federal Aviation Regulation No. 88	
	Fuel Tank System Fault Tolerance Evaluation Requirements	
Subpart A	General	21.1 – 21.9
Subpart B	Type Certificates	21.11 – 21.55
Subpart C	Provisional Type Certificates	21.71 – 21.85
Subpart D	Changes to Type Certificates	21.91 – 21.101
Subpart E	Supplemental Type Certificates	21.111 – 21.120
Subpart F	Production Under Type Certificate	21.121 – 21.130
Subpart G	Production Certificates	21.131 – 21.150
Subpart H	Airworthiness Certificates	21.171 – 21.199
Subpart I	Provisional Airworthiness Certificates	21.211 – 21.225
Subpart J [Reserved]		
Subpart K	Parts Manufacturer Approvals	21.301 – 21.320
Subpart L	Export Airworthiness Approvals	21.321 – 21.335
Subpart M [Reserved]		
Subpart N	Acceptance of Aircraft Engines, Propellers, and Articles for Import	21.500 – 21.502
Subpart O	Technical Standard Order Approvals	21.601 – 21.621
Subpart P	Special Federal Aviation Regulations	21.700

Delegates the airworthiness certification requirements of airworthiness to aircraft and operational type via additional Parts of Title 14 CFR:

- 14 CFR Part 23 (general aviation),
- 14 CFR Part 25 (transport category),
- 14 CFR Part 27 (general aviation rotorcraft), and
- 14 CFR Part 29 (transport category).

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Title 14 CFR Part 21

- Foundational Regulation for Airworthiness Requirement

- Aircraft designers seek a **Type Certificate (TC)** certifying the aircraft design is airworthy.
  - Entire aircraft certified under a TC
  - Onboard equipment cannot be modified under a type certificate.
  - Does not certify the individual components.

- **Supplementary type certificates (STCs)** can be issued for aircraft that are modifications to an existing TC'd aircraft.
  - Approves airworthiness
  - Operational limits of the modified aircraft

  - Airworthiness Certification determined based upon the safety case put forward by design engineers. Must meet hardware and software Design Assurance Levels.

Subpart A—General

§ 21.1 Applicability and definitions.

(a) This part prescribes—

(1) Procedural requirements for issuing and changing—

(i) Design approvals;

(ii) Production approvals;

(iii) Airworthiness certificates; and

(iv) Airworthiness approvals;

(2) Rules governing applicants for, and holders of, any approval or certificate specified in paragraph (a)(1) of this section; and

(3) Procedural requirements for the approval of articles.

(b) For the purposes of this part—

(1) ***Airworthiness approval*** means a document, issued by the FAA for an aircraft, aircraft engine, propeller, or article, which certifies that the aircraft, aircraft engine, propeller, or article conforms to its approved design and is in a condition for safe operation, unless otherwise specified;

(2) ***Article*** means a material, part, component, process, or appliance;

(3) ***Commercial part*** means an article that is listed on an FAA-approved Commercial Parts List included in a design approval holder's Instructions for Continued Airworthiness required by § 21.50;

(4) ***Design approval*** means a type certificate (including amended and supplemental type certificates) or the approved design under a PMA, TSO authorization, letter of TSO design approval, or other approved design;

(5) ***Interface component*** means an article that serves as a functional interface between an aircraft and an aircraft engine, an aircraft engine and a propeller, or an aircraft and a propeller. An interface component is designated by the holder of the type certificate or the supplemental type certificate who controls the approved design data for that article;

(6) ***Product*** means an aircraft, aircraft engine, or propeller;

(7) ***Production approval*** means a document issued by the FAA to a person that allows the production of a product or article in accordance with its approved design and approved quality system, and can take the form of a production certificate, a PMA, or a TSO authorization;

(8) ***State of Design*** means the country or jurisdiction having regulatory authority over the organization responsible for the design and continued airworthiness of a civil aeronautical product or article;

(9) ***State of Manufacture*** means the country or jurisdiction having regulatory authority over the organization responsible for the production and airworthiness of a civil aeronautical product or article.

(10) ***Supplier*** means a person at any tier in the supply chain who provides a product, article, or service that is used or consumed in the design or manufacture of, or installed on, a product or article.

Title 14 CFR Part 23

- Defines the airworthiness regulations of general aviation aircraft
  - Organized by:
    - Flight
    - Structures
    - Design and Construction
    - Powerplant
    - Equipment
    - Flight Crew Interfaces

▼ **Part 23** Airworthiness Standards: Normal Category Airplanes	23.1457 – 23.2620
§ 23.1457 Cockpit voice recorders.	
§ 23.1459 Flight data recorders.	
§ 23.1529 Instructions for continued airworthiness.	
▼ **Subpart A** General	23.2000 – 23.2010
§ 23.2000 Applicability and definitions.	
§ 23.2005 Certification of normal category airplanes.	
§ 23.2010 Accepted means of compliance.	
▶ **Subpart B** Flight	23.2100 – 23.2165
▶ **Subpart C** Structures	23.2200 – 23.2270
▶ **Subpart D** Design and Construction	23.2300 – 23.2335
▶ **Subpart E** Powerplant	23.2400 – 23.2440
▶ **Subpart F** Equipment	23.2500 – 23.2550
▼ **Subpart G** Flightcrew Interface and Other Information	23.2600 – 23.2620
§ 23.2600 Flightcrew interface.	
§ 23.2605 Installation and operation.	
§ 23.2610 Instrument markings, control markings, and placards.	
§ 23.2615 Flight, navigation, and powerplant instruments.	
§ 23.2620 Airplane flight manual.	
Appendix A to Part 23	
Instructions for Continued Airworthiness	

Subpart A—General

§ 23.2000 Applicability and definitions.

(a) This part prescribes airworthiness standards for the issuance of type certificates, and changes to those certificates, for airplanes in the normal category.

(b) For the purposes of this part, the following definition applies:

Continued safe flight and landing means an airplane is capable of continued controlled flight and landing, possibly using emergency procedures, without requiring exceptional pilot skill or strength. Upon landing, some airplane damage may occur as a result of a failure condition.

§ 23.2005 Certification of normal category airplanes.

(a) Certification in the normal category applies to airplanes with a passenger-seating configuration of 19 or less and a maximum certificated takeoff weight of 19,000 pounds or less.

(b) Airplane certification levels are:

(1) Level 1—for airplanes with a maximum seating configuration of 0 to 1 passengers.

(2) Level 2—for airplanes with a maximum seating configuration of 2 to 6 passengers.

(3) Level 3—for airplanes with a maximum seating configuration of 7 to 9 passengers.

(4) Level 4—for airplanes with a maximum seating configuration of 10 to 19 passengers.

(c) Airplane performance levels are:

(1) Low speed—for airplanes with a V_{NO} and $V_{MO} \leq 250$ Knots Calibrated Airspeed (KCAS) and a $M_{MO} \leq 0.6$.

(2) High speed—for airplanes with a V_{NO} or $V_{MO} > 250$ KCAS or a $M_{MO} > 0.6$.

(d) Airplanes not certified for aerobatics may be used to perform any maneuver incident to normal flying, including—

(1) Stalls (except whip stalls); and

(2) Lazy eights, chandelles, and steep turns, in which the angle of bank is not more than 60 degrees.

(e) Airplanes certified for aerobatics may be used to perform maneuvers without limitations, other than those limitations established under subpart G of this part.

§ 23.2010 Accepted means of compliance.

(a) An applicant must comply with this part using a means of compliance, which may include consensus standards, accepted by the Administrator.

(b) An applicant requesting acceptance of a means of compliance must provide the means of compliance to the FAA in a form and manner acceptable to the Administrator.

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Snippet from Part 23

Subpart A—General

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§23.2000 Applicability and definitions.

(a) This part prescribes airworthiness standards for the issuance of type certificates, and changes to those certificates, for airplanes in the normal category.

(b) For the purposes of this part, the following definition applies:

Continued safe flight and landing means an airplane is capable of continued controlled flight and landing, possibly using emergency procedures, without requiring exceptional pilot skill or strength. Upon landing, some airplane damage may occur as a result of a failure condition.

§23.2105 Performance data.

(a) Unless otherwise prescribed, an airplane must meet the performance requirements of this subpart in—

(1) Still air and standard atmospheric conditions at sea level for all airplanes; and

(2) Ambient atmospheric conditions within the operating envelope for levels 1 and 2 high-speed and levels 3 and 4 airplanes.

(b) Unless otherwise prescribed, the applicant must develop the performance data required by this subpart for the following conditions:

(1) Airport altitudes from sea level to 10,000 feet (3,048 meters); and

(2) Temperatures above and below standard day temperature that are within the range of operating limitations, if those temperatures could have a negative effect on performance.

(c) The procedures used for determining takeoff and landing distances must be executable consistently by pilots of average skill in atmospheric conditions expected to be encountered in service.

(d) Performance data determined in accordance with paragraph (b) of this section must account for losses due to atmospheric conditions, cooling needs, and other demands on power sources.

- 14 CFR § 23.2105(b)
  - Interpret as
    - Title 14 of Code of Federal Reg.
    - Part 23
    - Section 2105
    - Paragraph b
  - §§ is used if discussing multiple sections.

- Note: Part 23 was updated in 2017, which included a significant rework to be less prescriptive with design requirements and focus upon performance-based airworthiness standards.
  - i.e. what must be achieved, but not how.

Part 23 Snippets (2)

Subpart F—Equipment

§ 23.2500 Airplane level systems requirements.

This section applies generally to installed equipment and systems unless a section of this part imposes requirements for a specific piece of equipment, system, or systems.

(a) The equipment and systems required for an airplane to operate safely in the kinds of operations for which certification is requested (Day VFR, Night VFR, IFR) must be designed and installed to—

(1) Meet the level of safety applicable to the certification and performance level of the airplane; and

(2) Perform their intended function throughout the operating and environmental limits for which the airplane is certificated.

(b) The systems and equipment not covered by paragraph (a) of this section—considered separately and in relation to other systems—must be designed and installed so their operation does not have an adverse effect on the airplane or its occupants.

[Doc. No. FAA-2015-1621, Amdt. 23-64, 81 FR 96689, Dec. 30, 2016, as amended by Doc. No. FAA-2022-1355, Amdt. 23-65, 87 FR 75710, Dec. 9, 2022]

§ 21.24 Issuance of type certificate: primary category aircraft.

(a) The applicant is entitled to a type certificate for an aircraft in the primary category if—

(1) The aircraft—

(i) Is unpowered; is an airplane powered by a single, naturally aspirated engine with a 61-knot or less V_{so} stall speed as determined under part 23 of this chapter; or is a rotorcraft with a 6-pound per square foot main rotor disc loading limitation, under sea level standard day conditions;

(ii) Weighs not more than 2,700 pounds; or, for seaplanes, not more than 3,375 pounds;

(iii) Has a maximum seating capacity of not more than four persons, including the pilot; and

(iv) Has an unpressurized cabin.

(2) The applicant has submitted—

(i) Except as provided by paragraph (c) of this section, a statement, in a form and manner acceptable to the FAA, certifying that: the applicant has completed the engineering analysis necessary to demonstrate compliance with the applicable airworthiness requirements; the applicant has conducted appropriate flight, structural, propulsion, and systems tests necessary to show that the aircraft, its components, and its equipment are reliable and function properly; the type design complies with the airworthiness standards and noise requirements established for the aircraft under § 21.17(f); and no feature or characteristic makes it unsafe for its intended use;

(ii) The flight manual required by § 21.5(b), including any information required to be furnished by the applicable airworthiness standards;

(iii) Instructions for continued airworthiness in accordance with § 21.50(b); and

(iv) A report that: summarizes how compliance with each provision of the type certification basis was determined; lists the specific documents in which the type certification data information is provided; lists all necessary drawings and documents used to define the type design; and lists all the engineering reports on tests and computations that the applicant must retain and make available under § 21.49 to substantiate compliance with the applicable airworthiness standards.

(3) The FAA finds that—

(i) The aircraft complies with those applicable airworthiness requirements approved under § 21.17(f) of this part; and

(ii) The aircraft has no feature or characteristic that makes it unsafe for its intended use.

(b) An applicant may include a special inspection and preventive maintenance program as part of the aircraft's type design or supplemental type design.

(c) For aircraft manufactured outside of the United States in a country with which the United States has a bilateral airworthiness agreement for the acceptance of these aircraft, and from which the aircraft is to be imported into the United States—

(1) The statement required by paragraph (a)(2)(i) of this section must be made by the civil airworthiness authority of the exporting country; and

(2) The required manuals, placards, listings, instrument markings, and documents required by paragraphs (a) and (b) of this section must be submitted in English.

[Doc. No. 23345, 57 FR 41367, Sept. 9, 1992, as amended by Amdt. 21-75, 62 FR 62808, Nov. 25, 1997; Doc. No. FAA-2015-1621, Amdt. 21-100, 81 FR 96689, Dec. 30, 2016]

TSOs

- Technical Standard Orders (TSOs) are issued for standard components that are interchangeable such as altimeters.

- The performance standards are defined in the TSO and the equipment is marked with a TSO number.

Department of Transportation
Federal Aviation Administration
Aircraft Certification Service
Washington, DC

TSO-C10b

Date: 9/1/59

Technical Standard Order

Subject: TSO-C10b, ALTIMETER, PRESSURE ACTUATED, SENSITIVE TYPE

Part 514 contains minimum performance standards and specifications of materials parts, processes, and appliances used in aircraft and implements the provisions of sections 3.18, 4a.31, 4b.18, 6.18 and 7.18 of the Civil Air Regulations. The regulation uses the Technical Standard

SAE standard for Altimeters supports (Note: example from CAA, pre-FAA)

§ 91.225 Automatic Dependent Surveillance-Broadcast (ADS-B) Out equipment and use

(a) After January 1, 2020, unless otherwise authorized by ATC, no person may operate an aircraft in Class A airspace unless the aircraft has equipment installed that—

(1) Meets the performance requirements in—

(i) TSO-C166b and Section 2 of RTCA DO-260B (as referenced in TSO-C166b); or

(ii) TSO-C166c and Section 2 of RTCA DO-260C as modified by DO-260C—Change 1 (as referenced in TSO-C166c); and

(2) Meets the requirements of § 91.227.

(b) After January 1, 2020, except as prohibited in paragraph (h)(2) of this section or unless otherwise authorized by ATC, no person may operate an aircraft below 18,000 feet MSL and in airspace described in paragraph (d) of this section unless the aircraft has equipment installed that—

(1) Meets the performance requirements in—

(i) TSO-C166b and Section 2 of RTCA DO-260B (as referenced in TSO-C166b);

(ii) TSO-C166c and Section 2 of RTCA DO-260C as modified by DO-260C—Change 1 (as referenced in TSO-C166c);

(iii) TSO-C154c and Section 2 of RTCA DO-282B (as referenced in TSO-C154c); or

(iv) TSO-C154d and Section 2 of RTCA DO-282C (as referenced in TSO-C154d);

(2) Meets the requirements of § 91.227.

(c) Operators with equipment installed with an approved deviation under § 21.618 of this chapter also are in compliance with this section.

Technical Standard Order

Subject: Extended Squitter Automatic Dependent Surveillance - Broadcast (ADS-B) and Traffic Information Service - Broadcast (TIS-B) Equipment Operating on the Radio Frequency of 1090 Megahertz (MHz)

1. PURPOSE. This technical standard order (TSO) is for manufacturers applying for a TSO authorization (TSOA) or letter of TSO design approval (LODA). In it, we (the Federal Aviation Administration, (FAA)) tell you what minimum performance standards (MPS) your 1090 MHz ADS-B and TIS-B equipment must meet for approval and identification with the applicable TSO marking.

2. APPLICABILITY. This TSO affects new applications submitted after its effective date.

a. TSO-C166b will also remain effective until 18 months after the effective date of an amendment to Title 14, Code of Regulations (14 CFR) 91.225 and 91.227 to allow installation and use of ADS-B Out equipment manufactured under this TSO for compliance with those rules. After that date, we will no longer accept applications for TSO-C166b.

b. 1090 MHz ADS-B and TIS-B equipment approved under a previous TSOA may still be manufactured under the provisions of its original approval.

Note: 1090 MHz ADS-B and TIS-B equipment produced to TSO versions before TSO-C166b does not comply with the operating rule requirements of §§ 91.225 and 91.227.

3. REQUIREMENTS. New models of 1090 MHz ADS-B and TIS-B equipment identified and manufactured on or after the effective date of this TSO must meet the requirements in RTCA/DO-260C / EUROCAE ED-102B, *Minimum Operational Performance Standards for 1090 MHz Extended Squitter Automatic Dependent Surveillance-Broadcast (ADS-B) and Traffic Information Services-Broadcast (TIS-B)*, Sections 2.1 and 2.2, issued in December 2020, as modified by Change 1, issued in January 2022.

RTCA, Inc.
1150 18th Street, NW, Suite 910
Washington, DC 20036, USA

Minimum Operational Performance Standards for
1090 MHz Extended Squitter
Automatic Dependent Surveillance – Broadcast (ADS-B)
and
Traffic Information Services – Broadcast (TIS-B)

Change 1

RTCA DO-260C - Change 1
January 25, 2022

Prepared by: SC-186
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Airworthiness Directives

- Legally enforceable directives issued by the FAA to correct an unsafe condition in a product.
- Identifies the issue
- Defines the required procedure to address the issue
- Applicable toward aircraft, aircraft engines, propellers, or appliances

- Types include:
    - Notice of Public Rulemaking (NPRM),
        - Final Rule – Public Comment, and
    - Emergency NPRM

Battery Damage in the 787

Resulted in 2013 grounding of the Boeing 787

FAA
Aviation Safety

EMERGENCY AIRWORTHINESS DIRECTIVE

www.faa.gov/aircraft/safety/alerts/

DATE: January 16, 2013
AD #: 2013-02-51

Emergency airworthiness directive (AD) 2013-02-51 is sent to owners and operators of The Boeing Company Model 787-8 airplanes.

Background

This emergency AD was prompted by recent incidents involving lithium ion battery failures that resulted in release of flammable electrolytes, heat damage, and smoke on two Model 787-8 airplanes. The cause of these failures is currently under investigation. These conditions, if not corrected, could result in damage to critical systems and structures, and the potential for fire in the electrical compartment.

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2013-02-51 The Boeing Company: Directorate Identifier 2013-NM-016-AD.

(a) Effective Date

This Emergency AD is effective upon receipt.

(b) Affected ADs

None.

(c) Applicability

This AD applies to all The Boeing Company Model 787-8 airplanes, certificated in any category.

(d) Subject

Joint Aircraft System Component (JASC)/Air Transport Association (ATA) of America Code 24, Electrical power.

(e) Unsafe Condition

This AD was prompted by recent incidents involving lithium ion battery failures that resulted in release of flammable electrolytes, heat damage, and smoke on two Model 787-8 airplanes. The cause of these failures is currently under investigation. We are issuing this AD to prevent damage to critical systems and structures, and the potential for fire in the electrical compartment.

(f) Compliance

Comply with this AD within the compliance times specified, unless already done.

(g) Modification or Other Action

Before further flight, modify the battery system, or take other actions, in accordance with a method approved by the Manager, Seattle Aircraft Certification Office (ACO), FAA.

(h) Alternative Methods of Compliance (AMOCs)

(1) The Manager, Seattle ACO, FAA, has the authority to approve AMOCs for this AD, if requested using the procedures found in 14 CFR 39.19. In accordance with 14 CFR 39.19, send your request to your principal inspector or local Flight Standards District Office, as appropriate. If sending information directly to the manager of the ACO, send it to the attention of the person identified in the Related Information section of this AD. Information may be emailed to: 9-ANM-Seattle-ACO-AMOC-Requests@faa.gov.

(2) Before using any approved AMOC, notify your appropriate principal inspector, or lacking a principal inspector, the manager of the local flight standards district office/certificate holding district office.

Airworthiness directive's sections serve specific purposes:

- When and under what conditions is it applicable.
- Impacted component
- Safety issue
- Corrective action
- AMOC required if following a different approach to address the equipment and unsafe condition.

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Other Agencies

- ICAO, a United Nations Organization
  - Coordinates aviation standards across the world adopting and harmonizing standards and practices including navigation systems, infrastructure, aircraft identifiers, and many other regulations that need harmonization across borders.

- Civil Aviation Authorities (CAAs)
  - CAAs are the analogue to the FAA in other countries or regions throughout the world.
  - Examples:
    - Europe: European Aviation Safety Agency (EASA)
    - Australia: Civilization Aviation Safety Agency (CASA)

Aviation/Aerospace Standards

- **Standards Development Organizations** produce aviation and aerospace standards that address safety, reliability, and interoperability of aircraft/spacecraft systems.
  - Produce industry-supported consensus standards.

- **RTCA, Inc.** produce a variety of aviation standards used within this industry and relevant to this course.
  - e.g RTCA DO-178c: Airborne Software Certification defines standards to develop software to meet design assurance level based on system criticality and failure risk.

Other SDOs

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- **International Organization for Standards (ISO)** – operations, data, production, operation, and disposal
- **ASTM International** – materials used in spacecraft
- **SAE International** – space technologies, components, and materials
- **IEEE** – wired and wireless communication standards
- **Consultative Committee for Space Data Systems (CCSDS)** – data and communication protocols to ensure interoperability between organizations and nations.
- **American Institute of Aeronautics and Astronautics (AIAA)** – national aerospace standards and support of some ISO technical groups

Who Writes the Standards for Aviation Computing?

SAE International	System & safety processes for avionics	**ARP4754A/B** – aircraft & system development; **ARP4761/A** – safety assessment	Framework for designing & analysing complex avionics systems; defines safety levels and links system safety to software/hardware assurance.
RTCA + EUROCAE	Assurance for software, electronics, environment, data, security	**DO-178C / ED-12C** – airborne software; **DO-254 / ED-80** – electronic hardware; **DO-160 / ED-14** – environmental tests; **DO-278A / ED-109A** – ground CNS/ATM SW; **DO-200B / ED-76A** – aeronautical data; **DO-326A/356A** – cybersecurity; **DO-297** – Integrated Modular Avionics	Main "how-to" rulebooks. Regulators accept compliance to these as evidence that avionics software, hardware, data and security are safe enough for flight.
ARINC / AEEC (SAE ITC)	Interfaces & architectures for avionics computers & networks	**ARINC 429** – classic data bus; **ARINC 664 Pt 7** – AFDX network; **ARINC 653** – partitioned RTOS interface; **ARINC 661** – cockpit displays	Make avionics from different vendors interoperable: define message formats, deterministic networks, RTOS APIs and cockpit display interfaces.
ASTM (esp. F38)	Consensus rules for small UAS avionics & C2	**F3002** – design of command-and-control system for small UAS (plus related F38 standards)	Often used instead of the full DO-178C/DO-254 stack for small drones; gives practical requirements for autopilots, datalinks and fail-safe behaviour.
ISO / ASD-STAN	Component & test standards affecting avionics hardware	**ISO 2685** – fire tests for equipment in engine zones; various ISO/EN fastener, connector, wiring and material standards	Building blocks inside equipment specifications: define detailed part requirements and test methods that avionics hardware must satisfy.
ICAO (context)	High-level global rules for CNS, safety, environment	**Annex 10** – aeronautical telecommunications; **Annex 19** – safety management; others	Sets top-level global requirements for communication, navigation, surveillance and safety. National regulations and technical standards are written to satisfy these.

How the Standards Stack for Avionics & Computers

Layer in the stack	SDO & key standards	Role in avionics computing
System & safety framework	**SAE** – ARP4754A/B, ARP4761/A	Tell us how to design and analyse complex aircraft/avionics systems and assign safety levels that software, hardware and networks must meet.
Airborne software	**RTCA/EUROCAE** – DO-178C (+ supplements: DO-330, DO-331, DO-332, DO-333)	Rulebook for safety-critical airborne software: planning, requirements, coding, testing and showing evidence at different criticality levels.
Electronic hardware (FPGAs, boards)	**RTCA/EUROCAE** – DO-254	Hardware twin of DO-178C: how to specify, design and verify complex airborne electronic hardware used inside avionics boxes.
Platform & networks (IMA, OS, data bus)	**ARINC/AEEC** – ARINC 429, 664 (AFDX), 653, 661; **RTCA/EUROCAE** – DO-297	ARINC defines interfaces & behaviour of data buses, deterministic networks, RTOS APIs and displays; DO-297 explains how to certify shared “Integrated Modular Avionics” platforms built on those pieces.
Environment & physical robustness	**RTCA/EUROCAE** – DO-160; **ISO** – e.g., ISO 2685	Define the environmental and fire tests avionics hardware must pass (temperature, vibration, EMI, lightning, fire, etc.) before it can be installed on an aircraft.
Data, ground systems & cybersecurity	**RTCA/EUROCAE** – DO-200B (aeronautical data), DO-278A (ground CNS/ATM software), DO-326A/356A/355 (airworthiness security)	Make sure the data feeding avionics, the ground systems they talk to, and their cybersecurity posture are engineered and justified to an acceptable level.
Small UAS / drones (lighter regime)	**ASTM F38** – e.g., F3002 for C2 systems	Gives a more lightweight, consensus-based path for drone autopilots and command-and-control systems, especially for small UAS operations.

MOPS and MASPS

MASPS Minimum Aviation System Performance Standards

From RTCA's website

MASPS specify characteristics that are useful to designers, installers, manufacturers, service providers and users of systems intended for operational use within a defined airspace. Where the systems are global in nature, international applications are taken into consideration (often working with European Organisation for Civil Aviation Equipment (EUROCAE) and/or International Civil Aviation Organization (ICAO)). MASPS describe the system (subsystems / functions) and provide information needed to understand the rationale for system characteristics, operational goals, requirements and typical applications. Definitions and assumptions essential to proper understanding of MASPS are provided as well as minimum system test procedures to verify system performance compliance (e.g., end-to-end performance verification).

MOPS and MASPS

MOPS Minimum Operational Performance Standards

From RTCA's website

MOPS provide standards for specific equipment(s) useful to designers, manufacturers, installers and users of the equipment. The word "equipment" used in MOPS includes all components and units necessary for the system to properly perform its intended function(s). MOPS provide the information needed to understand the rationale for equipment characteristics and requirements stated, describe typical equipment applications and operational goals, and establish the basis for required performance under the standard. Definitions and assumptions essential to proper understanding are provided as well as installed equipment tests and operational performance characteristics for equipment installations.

MOPS and MASPS

	Minimum Aviation System Performance Standards (MASPS) for Traffic Information Service – Broadcast (TIS-B)
DO #:	**DO-286B**

Title:	**Minimum Operational Performance Standards (MOPS) for Helicopter Terrain Awareness and Warning System (HTAWS) Airborne Equipment**
DO #:	**DO-309**

Testing

Title:	**Environmental Conditions and Test Procedures for Airborne Equipment**
DO #:	**DO-160G**
Issued:	**12/08/2010**
Committee:	**SC-135**
Description:	DO-160G provides standard procedures and environmental test criteria for testing airborne equipment for the entire spectrum of aircraft from light general aviation aircraft and helicopters through the "jumbo jets" and SST categories of aircraft. The document includes 26 Sections and three Appendices. Examples of tests covered include vibration, power input, radio frequency susceptibility, lightning and electrostatic discharge. **Coordinated with EUROCAE, RTCA/DO-160G and EUROCAE/ED-14G are identically worded.** DO-160G is recognized by the International Organization for Standardization (ISO) as de facto international standard ISO- 7137.

6.3.1 Category A—Standard Humidity Environment

The test profile is shown graphically in Figure 6-1. The procedure shall be in accordance with the following steps:

Step 1: Install the test item in the test chamber, and ensure its configuration is representative of that used in actual service.

Step 2: Stabilize the test item at 38±2 °C and 85±4 % RH.

Step 3: Over a two-hour period, ±10 minutes, raise the chamber temperature to 50±2 °C and humidity to 95±4 % RH.

Step 4: Maintain the chamber temperature at 50±2 °C and humidity at 95±4 % RH for six hours minimum.

Step 5: During the next 16-hour period, ±15 minutes, decrease the temperature gradually to 38±2 °C or lower. During this period, keep the humidity as high as possible and do not allow it to fall below 85 % RH.

Step 6: Steps 3, 4 and 5 constitute a cycle. Repeat these steps until a total of two cycles (48 hours of exposure) have been completed.

Step 7: At the end of the exposure period, remove the equipment from the test chamber and drain off (do not wipe) any condensed moisture. Within one hour after the two cycles are completed, apply normal supply power and turn on the equipment. Allow 15 minutes maximum following the application of primary power for the equipment to warm up. For equipment that does not require electrical power for operation, warm up the equipment for 15 minutes maximum by the application of heat not to exceed the short-time operating high temperature test as required by applicable equipment categories. Immediately following the warm-up period, make such tests and measurements as are necessary to DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

Questions?

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